DPP — DEPARTMENTAL POLICY AND PROCEDURE

TITLE:	CAPILLARY BLOOD GAS SAMPLING
DEPARTMENT:	NEONATAL INTENSIVE CARE UNIT
APPLIES TO:	NICU PHYSICIANS AND NURSES

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	NAME	POSITION	SIGNATURE
PREPARED BY:	MS. KRISTINE P. MAGNETICO	UNIT QUALITY REPRESENTATIVE	
	MS. SHATHA MOHAMMED	NICU HEAD NURSE	
REVIEWED BY:	MEDICAL EXECUTIVE COMMITTEE		
	DR. ABDULKARIM IBRAHIM	HEAD OF NICU	
	MS. MERRY JOY H. GALLEBO	NURSING QUALITY COORDINATOR	
	MR. KHALID GHURMULLAH ALGHAMDI	QPS DIRECTOR	
APPROVED BY:	MS. HAIFA SAAD MISFER	DIRECTOR OF NURSING	
	DR. RAKAN KHALIL ALGHAMDI	AED FOR MEDICAL SERVICES	

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1. PURPOSE

1.1. To obtain a capillary blood gas that will provide accurate results with minimal discomfort and potential for injury and infection.

2. DEFINITION & ABBREVIATION

2.1. **Capillary Blood Gas (CBG)** – are obtained to assess adequacy of oxygenation and ventilation to infants who do not have arterial/venous line access.

2.2. **Hospital Information System (HIS)** – Hospital Information System is an element of health informatics that focuses mainly on the administrational needs of hospitals.

3. RESPONSIBILITY

3.1. **NICU Physicians (Consultant, Specialists and Residents)**

3.1.1. Responsible in providing a written order in the physician's order sheet thru Hospital Information System (HIS).

3.2. **NICU Nurse (Head Nurse, Charge Nurse and Staff Nurse)**

3.2.1. Responsible in obtaining blood gas through capillary route.

3.2.2. Relay CBG report to physician.

4. POLICY

4.1. CBG will be done after a written order from Physician.

4.2. Nurses must be aware of when and how to relay critical CBG report to Physician.

4.3. All CBG report must be documented in the patient's medical file.

4.4. Capillary blood gas samplings should not be performed at/or through the following sites

4.4.1. Presence of posterior curvature of the heel as the device may puncture the bone

4.4.2. The fingers of neonate, to avoid nerve damage

4.4.3. On previous puncture sites

4.4.4. Cyanotic or poorly perfused tissues

4.4.5. Localized areas of infection

5. PROCEDURE

5.1. The physician will write order for capillary blood gas sampling in the patient's medical file.

5.2. The nurse verify physician's order for CBG

5.3. Identify the patient correctly.

5.4. Perform hand hygiene according to protocol and wear gloves

5.5. Warm the newborn's heel for about 5 to 15 minutes before the procedure.

5.6. Cleanse the site with alcohol swab (lateral and medial posterior surface)

5.7. Puncture/lance the site and wipe away the first drop of blood.

5.8. Collect the blood holding the capillary tube horizontally.

5.9. Obtain a continuous flow to avoid clotting and make sure that no air bubbles present.

5.10. Give the CBG sample to a registered nurse to process in blood gas machine in the unit.

5.11. Apply pressure on the lanced site with cotton.

5.12. Obtain results and notify the physician of the capillary blood gas results.

5.13. Document the procedure and the capillary blood gas report in the patient's medical file.

6. MEASUREMENT

6.1. 100% of the NICU staff is knowledgeable and adhere to this policy and procedures.

7. REFERENCES /CROSS- REFERENCES

7.1. PMSH-DPP-NICU-0011(v3): Capillary Blood Sampling, 11/06/2023, p. 1-3

7.2. CBAHI Standards 3rd edition, 01/01/2016, NICU.10, NICU.12 p.126-127

7.3. MOH KSA Guidelines for Neonatal Care

8. ATTACHMENT

8.1. None

9. REVISION CLAUSE

9.1. Changes to the hospital policy and procedure template based from the Al-Baha Health Cluster logo, font face and design.